

Auteur: Wout Vanderborght
 Studierichting: Informatica
 Oefeningenreeks 2D nr. 10.12

Bereken de volgende limiet:

$$\lim_{x \rightarrow -\frac{1}{2}} \frac{-6x^2 - x + 1}{-10x^2 - x + 2} = \frac{0}{0}$$

Teller ontbinden:

$$\frac{-1}{2} \left| \begin{array}{cc|c} -6 & -1 & 1 \\ & 3 & -1 \\ \hline -6 & 2 & 0 \end{array} \right|$$

Noemer ontbinden:

$$\frac{-1}{2} \left| \begin{array}{cc|c} -10 & -1 & 2 \\ & 5 & -2 \\ \hline -10 & 4 & 0 \end{array} \right|$$

$$\Leftrightarrow \lim_{x \rightarrow -\frac{1}{2}} \frac{\cancel{(2x+1)}(-6x+2)}{\cancel{(2x+1)}(-10x+4)} = \frac{5}{9}$$

References